LAB 7: FUZZ TESTING FOR INPUT VALIDATION BUGS

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SECTION : D

1. Objective

The objective of this lab is to explore fuzz testing as a security and quality-assurance technique by identifying and fixing input-validation bugs in the provided Python application using the Hypothesis property-based testing library.

2. Project files and tools

- Application: self_storage.py
- Test file: test_self_storage.py
- Language: Python 3.13.13
- Libraries: Hypothesis 6.170.15 and pytest 9.2.0
- Execution environment: Python virtual environment (.venv)

3. Functions tested

parse_rental(text): parses a unit and rental duration separated by a colon, then returns the title-cased unit and floating-point duration.

duration_average(months_csv): parses comma-separated integer durations and returns their integer average.

payable_rent(text): parses rent and discount separated by a vertical bar, applies an 18% increase, subtracts the discount, and prevents a negative payable amount.

4. Fuzz-testing approach

Each function was tested with Hypothesis using the strategy st.text() | st.none(). This allows Hypothesis to generate arbitrary text values and None. The initial tests checked whether each function could process generated inputs without raising an exception. Hypothesis reported minimized examples that exposed input-validation failures.

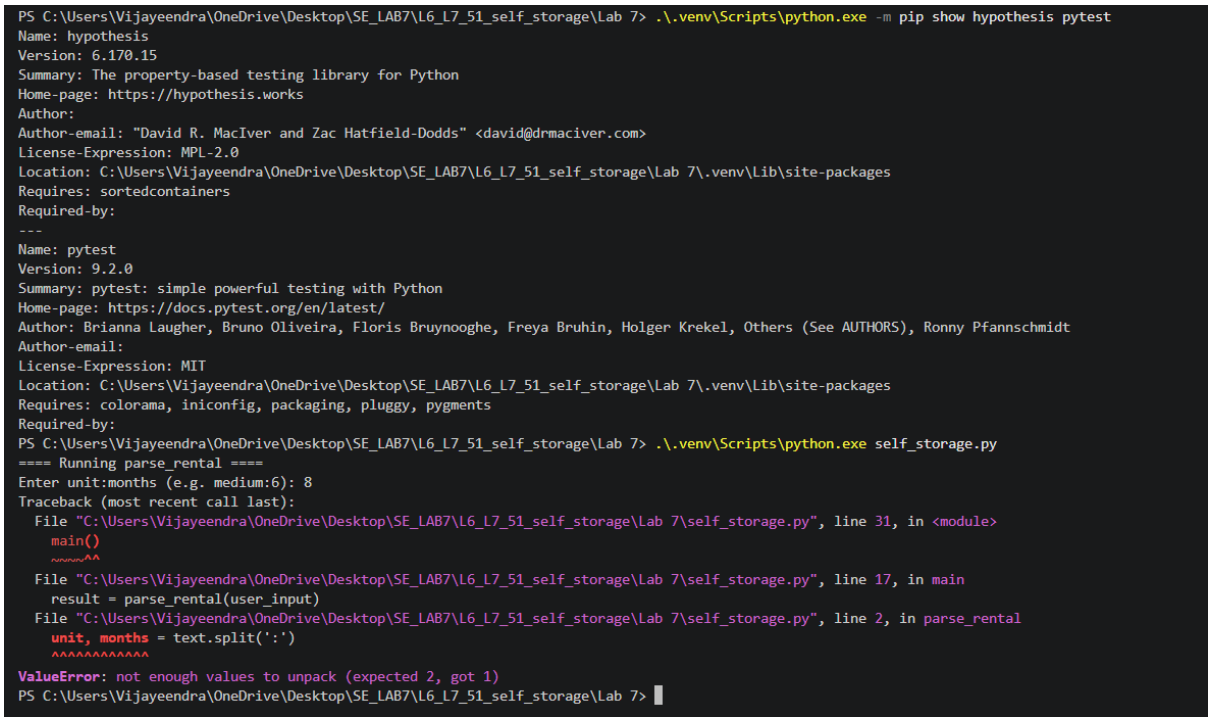
5. Bugs identified and corrective actions

Function	Observed failure examples	Correction
parse_rental	':', empty string, None	Validate input type, separator count, non-empty unit and finite non-negative duration; return ('Unknown', 0.0) for invalid input.

duration_average	empty string, None	Validate input type and non-empty CSV; parse each item as an integer and reject negative values; return 0 for invalid input.
payable_rent	'|', empty string, None	Validate input type and separator count; parse finite non-negative values and reject invalid calculations; return 0.0 for invalid input.

The fallback values above are the defensive behaviour implemented for this lab; the handout does not prescribe exact fallback values for invalid input.

6. Evidence screenshots





```
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\test_self_storage.py", line 30, in test_payable_rent_no_crash
    |     assert False, f"payable_rent crashed with input {s!r}: {e}"
    | AssertionError: payable_rent crashed with input '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Failing test case: test_payable_rent_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\test_self_storage.py", line 28, in test_payable_rent_no_crash
    |     payable_rent(s)
    |     ~~~~~~~~~~~~^^^
    |   File "C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\self_storage.py", line 10, in payable_rent
    |     rent, discount = text.split('|')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\test_self_storage.py", line 30, in test_payable_rent_no_crash
    |     assert False, f"payable_rent crashed with input {s!r}: {e}"
    | AssertionError: payable_rent crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Failing test case: test_payable_rent_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=========================== short test summary info ============================
FAILED test_self_storage.py::test_parse_rental_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_self_storage.py::test_duration_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_self_storage.py::test_payable_rent_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================== 3 failed in 10.22s ==============================
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7>
```

```
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7> .\.venv\Scripts\python.exe -m pytest test_self_storage.py -v
============================= test session starts =============================
platform win32 -- Python 3.13.13, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\.venv\Scripts\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7
plugins: hypothesis-6.170.15
collected 8 items

test_self_storage.py::test_parse_rental_no_crash PASSED                  [ 12%]
test_self_storage.py::test_duration_average_no_crash PASSED              [ 25%]
test_self_storage.py::test_payable_rent_no_crash PASSED                  [ 37%]
test_self_storage.py::test_parse_rental_valid PASSED                     [ 50%]
test_self_storage.py::test_duration_average_valid PASSED                 [ 62%]
test_self_storage.py::test_payable_rent_valid PASSED                     [ 75%]
test_self_storage.py::test_invalid_inputs PASSED                         [ 87%]
test_self_storage.py::test_non_finite_values PASSED                      [100%]

============================== 8 passed in 1.99s ==============================
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7>
```



```
============================== 8 passed in 1.99s ==============================
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7> .\.venv\Scripts\python.exe self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium:8
Unit and Months: ('Medium', 8.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3,6,12
Average Duration: 7

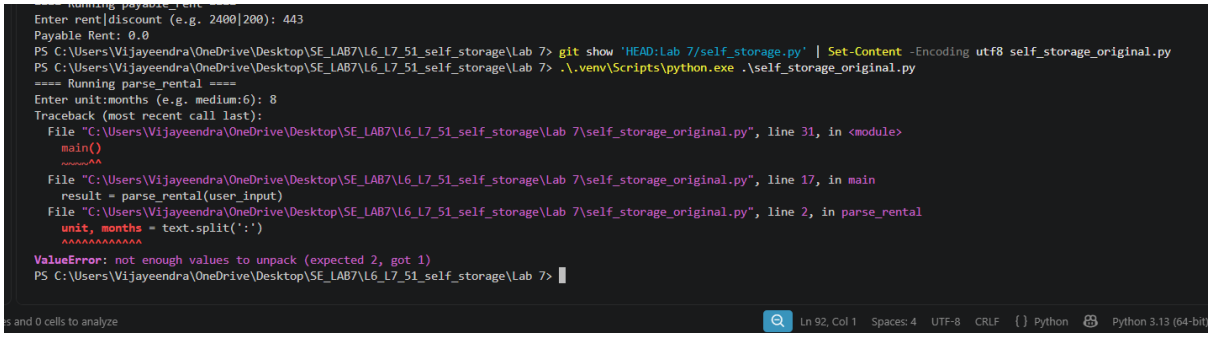
==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400|200
Payable Rent: 2632.0
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7>
```



```
Payable Rent: 2632.0
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7> .\.venv\Scripts\python.exe self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): "
Unit and Months: ('Unknown', 0.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3.4.5.
Average Duration: 0

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 443
Payable Rent: 0.0
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7>
```


Enter rent|discount (e.g. 2400|200): 443
Payable Rent: 0.0
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7> git show 'HEAD:Lab 7/self_storage.py' | Set-Content -Encoding utf8 self_storage_original.py
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7> .\.venv\Scripts\python.exe .\self_storage_original.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): 8
Traceback (most recent call last):
File "C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\self_storage_original.py", line 31, in <module>
main()
File "C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\self_storage_original.py", line 17, in main
result = parse_rental(user_input)
File "C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7\self_storage_original.py", line 2, in parse_rental
unit, months = text.split(':')
ValueError: not enough values to unpack (expected 2, got 1)
PS C:\Users\Vijayeendra\OneDrive\Desktop\SE_LAB7\L6_L7_51_self_storage\Lab 7>
and 0 cells to analyze
Ln 92, Col 1
Spaces: 4
UTF-8
CRLF
Python
Python 3.13 (64-bit)